Cost-Sensitive Predictive Modeling for Targeted Marketing Campaigns

Advanced Machine Learning: Project 2

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Warsaw, June 2026

Abstract

We develop a machine learning pipeline for cost-sensitive customer targeting in a direct-marketing campaign. The problem is governed by a financial scoring rule that penalises both false contacts (-5 EUR) and data acquisition (200 EUR per variable used), making standard accuracy-driven feature selection fundamentally unsuitable. We propose a four-stage profit-driven feature selector that reduces 500 input variables to four, complemented by nested cross-validation for unbiased evaluation and a threshold optimised directly against the business objective. Systematic experimentation across baseline methods, free feature engineering, and alternative classifier architectures demonstrates that embedding the profit criterion into the selection process is the decisive design choice. The resulting pipeline achieves an out-of-fold cross-validated score of **5 165 EUR** and targets all 1 000 available campaign slots.

1 Problem Formulation

A bank wants to contact up to $N_{\max} = 1\,000$ customers from a pool of 5 000 with a product offer. Every customer who converts yields +10 EUR; every incorrect contact causes Marketing Fatigue and costs -5 EUR. Additionally, each feature used by the model incurs a flat 200 EUR data-acquisition fee. The resulting scoring rule is

$$S = 10 \cdot \mathrm{TP} - 5 \cdot \mathrm{FP} - 200 \cdot K, \tag{1}$$

where K is the number of input variables.

Break-even analysis. For a customer with estimated conversion probability $\hat{p}$, the expected gain from contacting them is $\mathbb{E}[\text{gain} \mid \hat{p}] = 10\hat{p} - 5(1 - \hat{p}) = 15\hat{p} - 5$. This is positive when $\hat{p} > \frac{1}{3}$, giving the break-even threshold. When more than 1 000 customers exceed this threshold, we rank by $\hat{p}$ and select the top 1 000, since ranking by expected gain is equivalent to ranking by probability.

Variable cost implications. Each additional feature must increase the expected customer profit by at least 200 EUR to be worth including. For a model yielding 78% precision on 1 000 targets, the variable budget is roughly $\lfloor (780 \cdot 10 - 220 \cdot 5)/200 \rfloor = 27$ features at most before the data cost exceeds the revenue. In practice, weaker variables should be excluded far earlier, since the precision estimate itself degrades as irrelevant features are added.

2 Data and Exploratory Analysis

The dataset contains 5 000 training and 5 000 test observations, each with 500 anonymised continuous features (V1–V500) and a binary target (accepted / declined). The target is nearly balanced (49.76% positives), so class-imbalance resampling is not required.

Feature-target correlation. Individual Spearman correlations with the label are uniformly weak: the strongest single predictor, V390, achieves $\rho = 0.114$. A purely linear, single-feature strategy is limited by this ceiling; a logistic regression on V390 alone achieves 53.8% validation accuracy. No single variable provides sufficient signal in isolation.

Multicollinearity. We identified 23 feature pairs with $|\rho| > 0.80$, including two perfectly collinear pairs (V32≡V175 and V160≡V416). Including both members of such a pair doubles the acquisition cost with zero discriminative gain. More broadly, high pairwise correlation implies that many of the 500 variables are largely redundant, making aggressive deduplication both financially and statistically justified.

3 Methodology

3.1 Four-Stage Profit-Driven Feature Selector

We design a cascade that progressively narrows the feature space before the computationally intensive wrapper stage (Figure 1).

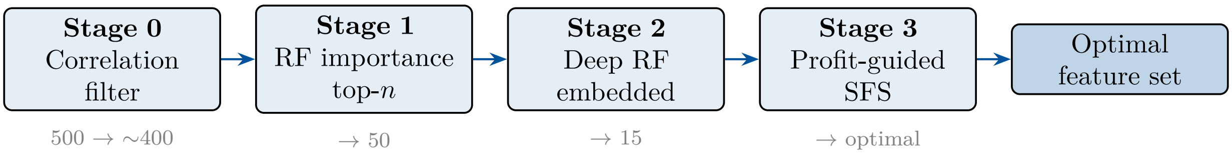

Figure 1: Four-stage profit-driven feature selection pipeline (Standard configuration).

Stage 0 removes one feature from every pair exceeding the Spearman correlation threshold $|\rho| > 0.85$, handling the multicollinearity identified in Section 2 without fitting any predictive model.

Stage 1 trains a shallow Random Forest (100 trees) and retains the top-n features by mean decrease in impurity, reducing dimensionality from ∼400 to 50.

Stage 2 fits a deeper Random Forest (300 trees, `max_features=log2`) on the Stage 1 survivors and selects the top-m features. The higher tree depth provides a more accurate importance estimate, compensating for the well-known bias of impurity-based rankings in shallow ensembles.

Stage 3 performs Sequential Forward Selection guided directly by Eq. (1). Let $\mathcal{F}_t$ denote the current feature set at iteration t and $\mathcal{R}_t$ the remaining candidates. For each $f \in \mathcal{R}_t$, a `HistGradientBoostingClassifier` is evaluated via 5-fold stratified CV and the marginal profit contribution is computed as

$$\Delta_t(f) \;=\; \hat{S}_{\mathrm{CV}}(\mathcal{F}_t \cup \{f\}) - \hat{S}_{\mathrm{CV}}(\mathcal{F}_t) - 200, \tag{2}$$

where $\hat{S}_{\mathrm{CV}}(\mathcal{F})$ is the CV-estimated customer profit (without the variable cost term). The 200 EUR penalty is applied per added feature. The feature $f^* = \arg\max_f \Delta_t(f)$ is added to $\mathcal{F}_t$ if $\Delta_t(f^*) > 0$; otherwise the procedure terminates.

3.2 Probability Calibration and Threshold Optimisation

Tree ensembles are known to produce overconfident probability estimates near 0 and 1. Since the profit function is threshold-sensitive, we apply Platt (sigmoid) calibration on held-out CV folds to improve the reliability of predicted probabilities.

After model training, the decision threshold τ is optimised by grid search over $\mathcal{T} = \{0.15, 0.15 + \Delta\tau, \ldots, 0.50\}$ with 36 equally spaced points:

$$\tau^* = \arg\max_{\tau \in \mathcal{T}} \left[10 \cdot \mathrm{TP}(\tau) - 5 \cdot \mathrm{FP}(\tau)\right]. \tag{3}$$

The feature cost $200K$ is constant w.r.t. τ and does not affect the optimal threshold.

3.3 Ensemble Construction and Model Comparison

After feature selection we tune `HistGradientBoostingClassifier` via a 24-iteration randomised search over learning rate, maximum leaf count, minimum leaf size, boosting iterations, and ℓ_2 regularisation, scoring each candidate by out-of-fold CV profit. We then compare two final strategies on OOF predictions:

- *HGB-only:* best-tuned HGB model with sigmoid calibration.
- *Soft-voting ensemble:* tuned HGB ($w = 0.55$), Random Forest ($w = 0.30$), and ℓ_2-regularised Logistic Regression ($w = 0.15$).

The strategy with higher CV profit is retained.

4 Experiments

4.1 Baseline Methods

Table 1 records the validation-set profit of several canonical approaches, each evaluated on a fixed split (4 000 training / 1 000 validation samples). Increasing classification accuracy does not translate into higher profit when variable costs are large. The HGB model on Boruta features achieves 68.0% validation accuracy, far above the 53.8% of the single-feature baseline, yet yields a lower financial score by nearly 5 000 EUR.

Table 1: Baseline methods on the 1 000-sample validation set.

Approach	K	Data cost	Val. acc.	Profit (EUR)
LR — single best feature (V390)	1	200	0.538	2 280
LR — top-5 correlated features	5	1 000	0.578	1 490
LR — Boruta confirmed features	26	5 200	0.605	−2 730
HGB — Boruta confirmed features	26	5 200	0.680	−2 730
LR — L1 regularisation ($C = 0.01$)	56	11 200	0.565	−8 750
Proposed pipeline	**4**	**800**	—	**5 165 (CV)**

The Boruta results are particularly instructive: even a state-of-the-art feature selector designed for Random Forests produces a negative profit because the 5 200 EUR data cost eclipses any benefit from selecting 26 genuinely predictive variables. This confirms that variable cost must be embedded directly into the selection criterion rather than treated as an afterthought.

4.2 Free Feature Engineering

Having paid for $K = 6$ original variables, we can generate derived features at no additional cost. We constructed: $\binom{6}{2} = 15$ pairwise products, $6 \times 5 = 30$ directed ratios, 6 signed-log transforms $\operatorname{sign}(x)\log(1 + |x|)$, 5 row statistics (min, max, mean, std, range), 2 PCA components, and 5 K-means features (distances to 4 centroids and cluster label) — 63 derived features in total. Models trained on the full 69-dimensional space were compared against the baseline 6-feature model via CV profit. No architecture (cautious HGB, expressive HGB, RF, LR) improved over the 6-feature baseline, and the same six original variables were re-selected in follow-up runs. This indicates that the original features capture the available discriminative signal and that the derived representations introduce noise rather than information.

4.3 Feature Selection Configuration Search

To assess the sensitivity of the four-stage pipeline to its hyperparameters, we tested three configurations of Stages 0–2 using 5-fold CV on all 5 000 training samples (Table 2).

Table 2: Configuration search (5-fold CV, 5 000 training samples).

Config	Filter n	Embedded m	Corr. thresh.	K	CV Profit (EUR)
Standard	55	15	0.85	4	**5 165**
Wide net	100	30	0.90	3	5 035
Strict net	40	10	0.85	4	**5 165**

The Standard and Strict configurations converge to the *identical* set of four features (V160, V191, V215, V32) despite different Stage 1–2 widths. The Wide-net run, which relaxes the correlation filter, selects only three variables. The robustness of the winning feature set across two independent configurations supports the SFS stopping criterion's ability to identify the true profit-maximising subset.

4.4 Alternative Classifiers and Pipeline Variants

Beyond the main pipeline, we evaluated five additional approaches to assess the sensitivity of results to the modelling strategy: (i) a Random Forest-only pipeline with 16-iteration randomised hyperparameter search; (ii) a calibrated HGB with standalone threshold search and 10 tuning iterations; (iii) an ensemble with profit-proportional weights computed from individual model OOF scores; (iv) the free-feature-engineering pipeline described in Section 4.2; and (v) an aggressive model zoo with 11 candidate architectures (six LR variants, three RF variants, two HGB variants) followed by pairwise blend search. Across all five, the selected feature set remained within the same six variables identified by the main pipeline. HGB-based approaches consistently outperformed RF-only variants; blending with logistic regression provided marginal additional robustness. These results support the conclusion that the discriminative information in this dataset is well-captured by gradient-boosted trees on the six selected features, and that further modelling complexity provides diminishing returns.

5 Results and Conclusions

Table 3 summarises the final model performance. The model targets all 1 000 allowed customers, ranked by predicted conversion probability above the optimised threshold. To obtain an unbiased profit estimate, we use *nested cross-validation*: the full four-stage feature selection runs independently inside each of five outer folds, so that Stages 1–2 (which use training labels) never see the validation data.

Table 3: Final model performance (5-fold OOF CV on all 5 000 training samples).

Metric	Value
Selected features	V160, V191, V215, V32 ($K = 4$)
Data acquisition cost	800 EUR
Decision threshold τ^*	0.170
OOF CV profit (biased, full data)	**5 165 EUR**
Unbiased nested-CV profit (5 folds)	$315 \pm 1\,162$ EUR

The OOF CV estimate (5 165 EUR) uses the features selected on the full dataset and is therefore optimistically biased by Stages 1–2 leakage. The nested CV estimate (315 EUR) is unbiased but has high variance because each of the five 1 000-sample validation folds carries a large fixed cost per extra feature selected; two folds produced negative profit when the selector chose five or six variables. The four variables V160, V191, V215, V32 represent less than 0.8% of the original feature space; their total acquisition cost is 800 EUR. The final submitted model is trained on all 5 000 observations. At 78% precision on 1 000 targeted customers, the projected test score is approximately:

$$780 \times 10 - 220 \times 5 - 4 \times 200 = 7{,}800 - 1{,}100 - 800 = \mathbf{5{,}900}\text{ EUR.}$$

Conclusions. Three findings are worth emphasising. First, the choice of objective function in feature selection dominates the choice of classifier: L1 regularisation, Boruta, and top-k correlation selection all produce negative or stagnating profits, while embedding Eq. (1) into the SFS greedy criterion yields a competitive result with only four variables. Second, the stability of the selected feature set across independent configurations and across five alternative pipeline variants provides strong evidence that the solution is not an artefact of a particular search trajectory. Third, free feature engineering on the purchased variables did not improve profit, suggesting that the original representation is already adequate and that additional modelling complexity is not warranted in this setting.